

SeCoR: Evidence-Guided Selective Correction for Lightweight Remote Sensing Change Detection

Chaoyun Mai^{ID}, Haipeng He^{ID}, Hao Xie^{ID}, Yikui Zhai^{ID}, *Senior Member, IEEE*, Zhiyuan Su^{ID}, Panrong Chen, and Zhiping Peng

Abstract—Remote sensing change detection (RSCD) aims to identify land-cover changes from bitemporal images and is widely used in urban monitoring, disaster assessment, and land-resource analysis. Lightweight RSCD models usually rely on compact backbones and efficient temporal interaction, but most of them update bitemporal features in a dense or implicit manner, without explicitly verifying whether opposite-temporal evidence is reliable. This may amplify pseudochanges under appearance disturbances or miss weak true changes when valid support is limited. To address this issue, we propose SeCoR, an evidence-guided selective correction framework for lightweight RSCD. In the encoder, reliability-aware local support correction selectively aggregates reliable local support from the opposite temporal branch to correct features before fusion. In the decoder, adaptive prototype-prior repair converts confident coarse priors into image-specific prototypes to repair ambiguous low-level features. Experiments on four public optical RSCD benchmarks demonstrate a strong accuracy-efficiency tradeoff with only 2.50 M parameters and 2.66G FLOPs. SeCoR obtains the highest F1 and IoU scores on all four datasets, with only marginal gains on LEVIR-CD and larger gains on WHU-CD, SYSU-CD, and UAV-CD, while remaining competitive with the strongest method on LEVIR-CD. Five-seed evaluations further provide statistical evidence of reproducible improvements over the architecture-matched Baseline. A half-width variant reduces the model to 0.99M parameters and 1.99G FLOPs while retaining competitive accuracy.

Index Terms—Evidence-guided selective correction, lightweight neural networks, prototype-prior repair, reliability-aware local support correction, remote sensing change detection.

I. INTRODUCTION

REMOTE sensing change detection (RSCD) aims to identify land-cover changes from bitemporal remote sensing images and has been widely used in urban monitoring, disaster assessment, land-resource management, and infrastructure inspection [1], [2]. Surveys of RSCD show that the field has progressed from classical pixel- and object-based analysis to deep

Received 31 May 2026; revised 19 July 2026; accepted 2 August 2026. Date of publication 11 August 2026; date of current version 27 August 2026. This work was supported in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2025A1515110259, in part by the National Natural Science Foundation of China under Grant 62273109, and in part by the Guangdong Provincial Science and Technology Achievements Transfer Program under Grant 2026A0202020030. (Corresponding author: Hao Xie.)

The authors are with the College of Electronics and Information Engineering, Wuyi University, Jiangmen 529020, China (e-mail: maichaoyun@foxmail.com; skyitn1@163.com; xiehao_wyu@163.com; yikuizhai@163.com; zhiyuansu_504@163.com; 2289262543@qq.com; pzhp@wyu.edu.cn).

Our code is available at <https://github.com/haicyonn/SeCoR-Net>.

Digital Object Identifier 10.1109/JSTARS.2026.3721954

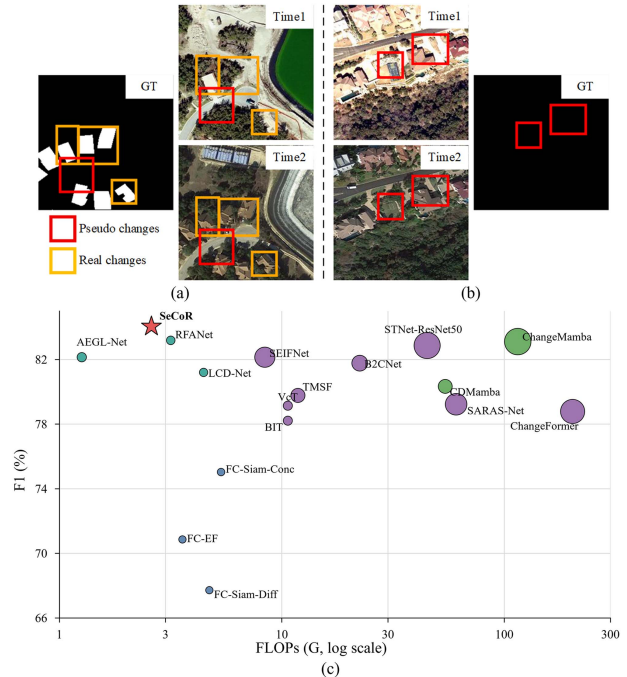


Fig. 1. Motivation and accuracy-efficiency overview. (a) Low-evidence true changes. (b) Appearance-induced pseudo changes.

dense prediction [3], [4]. Deep neural networks have advanced change prediction by learning discriminative spatial and semantic representations from image pairs [5], [6]. However, many high-performing RSCD models rely on heavy backbones, dense temporal interaction, or complex decoder refinement [7], [8], making the design of lightweight RSCD models an important problem for practical applications.

The main difficulty of lightweight RSCD lies in the limited representation capacity of compact models. As illustrated in Fig. 1(a) and (b), compact networks must handle two opposite ambiguity patterns: low-evidence true changes and appearance-induced pseudochanges [10], [11]. The key question is, therefore, not merely how to fuse more temporal evidence, but how to decide, which evidence is reliable enough to correct compact features. Compared with larger networks, lightweight models are more easily affected by unreliable visual evidence. Typical disturbances include pseudochanges caused by illumination, seasonal variation, moving objects, local misregistration, and viewpoint changes [10], [12], [13]. At the same time, true changes may be weak when the changed objects are small,

low-contrast, partially occluded, or located near cluttered backgrounds [36]. These factors may cause compact models to amplify irrelevant temporal differences or suppress subtle changed regions, leading to false alarms, missed detections, and inaccurate boundaries. Fig. 1(c) summarizes the accuracy-efficiency comparison on the SYSU-CD dataset, where SeCoR is compared with representative RSCD models in terms of detection accuracy and model complexity.

Existing RSCD methods address these challenges from different perspectives. CNN-based and lightweight methods reduce computation using compact backbones, feature differencing, concatenation, progressive aggregation, channel splitting, or lightweight attention designs [18], [19], [38]. Transformer- and attention-based methods strengthen temporal interaction and long-range dependency modeling, but they often increase computational cost [8], [24], [47]. State-space models provide another efficient sequence-modeling route for long-range change representation [27], [28], [29]. Multiscale and edge-aware methods recover details around changed regions [15], [36]. Alignment and frequency-domain methods reduce appearance disturbances [12], [13], [41]. However, these strategies mainly focus on making temporal features more expressive or densely fused. The key limitation is therefore not the absence of temporal interaction, but the lack of an explicit decision mechanism for determining whether local cross-temporal evidence should be trusted before compact features are updated.

This article is motivated by two observations. First, not all cross-temporal evidence should be fused. Some local candidates provide useful support for real changed regions, whereas others introduce pseudochange noise caused by misregistration or appearance variation [10], [13], [20].

Therefore, compact encoder features should be corrected only after reliable local support has been selected and verified.

Second, coarse predictions provide useful semantic guidance, but directly injecting them into low-level decoder features may over-smooth boundaries or propagate coarse errors. Inspired by prototype-based dense prediction [42], [57], a coarse prior can be more reliably used by converting its confident regions into image-specific semantic references, while ambiguous low-level details are repaired through a controlled residual path.

Motivated by these observations, we propose SeCoR, a lightweight evidence-guided selective correction framework. SeCoR asks when feature correction should be applied and which evidence should guide it under a lightweight budget. It contains two serially coupled but mechanistically distinct stages: RLSC performs prefusion encoder correction with reliability-checked local support, while adaptive prototype-prior repair (APPR) converts confident coarse-prior regions into image-specific prototype references for low-level decoder repair. The two stages are connected through the shared representation and coarse prior, without sharing an explicit confidence or routing variable.

The main contributions of this article are summarized as follows.

- 1) We propose SeCoR, an evidence-guided selective correction framework for lightweight RSCD. Instead of relying on dense temporal fusion or heavy global attention, SeCoR applies lightweight residual updates guided by reliability-checked local support and image-specific prototype priors.

- 2) We design reliability-aware local support correction (RLSC) for prefusion encoder correction. Rather than directly recalibrating or fusing available bitemporal features, RLSC includes explicit nonsupport entries in the routing softmax, aggregates only the selected opposite-temporal candidates, and gates the residual update with a confidence map that combines support availability and feature-support consistency.
- 3) We design APPR for decoder-side repair. Rather than using prototypes for output classification, APPR constructs a pair of image-specific changed and unchanged prototypes from the shared bitemporal representation using spatial weights derived from the coarse prior and its confidence, and uses the pair as an internal semantic reference for residual repair at designated low-level decoder stages.
- 4) Extensive experiments, controlled ablations, boundary evaluation, diagnostic analyses, and backbone-sensitivity studies demonstrate that SeCoR provides a strong accuracy-efficiency tradeoff with 2.50M parameters and 2.66G FLOPs. Five-seed evaluations show reproducible improvements over the architecture-matched Baseline, while a half-width variant provides a more compact alternative with 0.99M parameters and 1.99G FLOPs and retains competitive accuracy.

II. RELATED WORK

A. Deep Remote Sensing Change Detection

Traditional RSCD methods generate difference images through image differencing, image ratioing, change vector analysis, or transformation-based techniques, followed by thresholding or classification [30], [37], [49], [50]. These methods are simple but sensitive to radiometric variation, registration errors, and complex backgrounds [10], [11]. With deep learning, RSCD has been formulated as end-to-end dense prediction [33], [51]. Fully convolutional Siamese variants, including FC-EF, FC-Siam-conc, and FC-Siam-diff, established representative bi-temporal comparison baselines [5]. Dual-task constrained Siamese learning further showed the value of auxiliary supervision for building change detection [6].

Transformer-based and hybrid models further improve global context modeling and long-range temporal interaction. Nonlocal neural networks highlighted the importance of long-range dependency modeling in visual recognition [54]. Vision Transformer and Swin Transformer designs motivated attention-based dense recognition with global or hierarchical token interactions [22], [23]. BIT introduced semantic tokens for efficient bitemporal interaction in RSCD [8]. ChangeFormer exploited hierarchical Transformer representations for multiscale change decoding [47]. VcT introduced visual change tokens for remote sensing image CD [24]. STNet models temporal feature fusion with cross-temporal gating and spatial feature fusion with cross-scale attention [25]. SARAS-Net emphasizes scale and relation awareness in a Siamese framework [26]. TMSF uses a Transformer-based multiscale Siamese design with dual-attention tokens and adaptive multiscale fusion to combine local, global, and multiscale information [40]. Recent state-space models provide another route to efficient long-range modeling. Mamba introduces selective state spaces for linear-time sequence modeling

[27]. ChangeMamba uses a spatiotemporal state-space model for RSCD [28]. CDMamba incorporates local clues into Mamba for change detection [29].

These methods improve how temporal features are represented, aligned, or fused, but the update decision is usually embedded in dense fusion, attention, or sequence modeling. As a result, they seldom explicitly verify whether local cross-temporal evidence should be used for feature correction or suppressed as disturbance. This remains insufficiently explored in current RSCD methods.

B. Lightweight Change Detection

Lightweight RSCD is important for practical deployment because change detection is often performed on large-area remote sensing images through dense patchwise inference. In such scenarios, parameter number, FLOPs, memory usage, and inference speed directly affect the cost of large-scale mapping and time-sensitive applications. Existing lightweight designs reduce complexity through efficient backbones, compact decoders, feature reuse, channel reduction, depthwise separable convolutions, lightweight attention, or gating operations. MobileNetV2 and EfficientNet are commonly adopted as efficient CNN backbones, relying on inverted residuals with linear bottlenecks and compound scaling, respectively [18], [21], and channel or convolutional attention provides useful ideas for compact feature recalibration [55], [56].

Recent lightweight CD methods have moved beyond backbone replacement and introduced task-specific temporal interaction and refinement designs. AEGL-Net uses adaptive multiscale enhancement and global-local feature fusion to improve bitemporal feature representation [48]. LCD-Net uses temporal interaction and fusion, multiscale feature fusion, and decoder gating [38]. RFANet reinforces weak lightweight features through robust multilevel feature aggregation in a compact network [39]. Lighter further designs full-information mingling and dual-guide difference capture to strengthen compact temporal interaction under a lightweight budget [35].

Although these lightweight models differ in implementation, they use differencing, recalibration, gating, or feature fusion to combine information from the two temporal streams. RLSC instead separates correction from fusion. Before temporal fusion, it searches a local neighborhood in the opposite-temporal branch and uses a routing distribution that also contains explicit no-support entries. Probability assigned to these entries is excluded from support aggregation, and the residual update is determined jointly by the probability assigned to the selected local candidates and the agreement between their aggregate and the current feature.

C. Cross-Temporal Interaction and Prior-Guided Refinement

Pseudochange is a central difficulty in high-resolution RSCD [10], [20]. It can be caused by illumination variation, seasonal appearance changes, shadows, moving objects, local misregistration, or heterogeneous object appearance. CFNet

reduces style-induced disturbance through content-aware enhancement [16]. BiFA explicitly considers channel-level temporal interaction and spatial alignment to mitigate illumination and perspective-induced disturbances [13]. RACDNet uses deformable feature alignment to address resolution and registration inconsistencies in optical RSCD [34]. DCCANet introduces frequency-spatial collaborative modeling to suppress environmental interference and noise [41]. SEIFNet introduces spatiotemporal enhancement and interlevel fusion to improve change representation under scale variation and pseudochange interference [17]. These methods are effective for reducing illumination, viewpoint, and geometric disturbances, but they mainly focus on making features more comparable before fusion. In lightweight RSCD, however, compact features may still contain ambiguous local evidence after alignment or attention-based interaction. This motivates our focus on whether local cross-temporal evidence should support feature correction or be suppressed as disturbance.

Prior-guided and boundary-aware refinement methods address another aspect of the problem. High-level features provide semantic information but are spatially coarse, whereas low-level features preserve edges and fine structures but are sensitive to irrelevant texture. Feature pyramid and dilated-context designs show that multiscale representation is important for dense prediction [52], [53]. Edge supervision and boundary-aware modeling provide useful cues for precise localization [31]. MEDS-CD supplements high-frequency details for building boundary recovery [36]. Adaptive edge-guided fusion and EDGE-Net explicitly use edge information to enhance change localization [32], [15]. B2CNet performs boundary-to-center refinement for multitemporal change extraction [14]. Prototype-based dense prediction provides another useful clue by representing class-specific feature centers.

Many prior-guided refinement methods directly inject coarse predictions or high-level semantic priors into decoder features, which may propagate coarse errors or produce oversmoothed responses around changed boundaries. APPR instead constructs image-specific changed and unchanged prototypes from confident coarse-prior regions in the shared bitemporal representation and uses them as internal semantic references for low-level repair, rather than as an output classifier or alignment target. Thus, APPR differs in the role assigned to prototypes: they guide an internal residual repair path instead of producing the final prediction.

III. METHODOLOGY

This section presents the proposed network for lightweight RSCD. We first describe the overall architecture and the data flow of the two correction stages. Then, RLSC and APPR are introduced in detail, followed by the prediction logits and loss function.

A. Overall Architecture

Given two bitemporal remote sensing images T_1 and T_2 acquired in the same area, binary RSCD aims to predict a final map $\hat{Y}$. As shown in Fig. 2, SeCoR consists of a lightweight

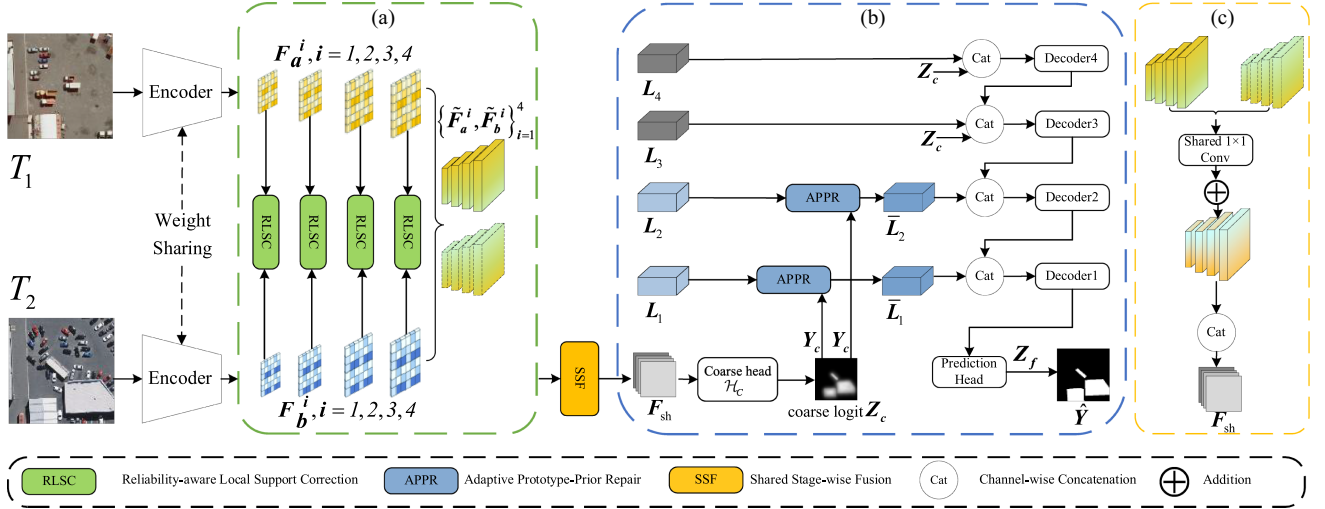


Fig. 2. Overall architecture of SeCoR. A shared MobileNetV2 encoder extracts multiscale bitemporal features from the input image pair. RLSC performs reliability-aware encoder correction before temporal fusion. SSF fuses the corrected bi-temporal features into a shared representation, from which a lightweight coarse head predicts a coarse prior. APPR then repairs ambiguous low-level decoder features using the shared representation and the coarse prior. The decoder aggregates the repaired low-level features and the remaining multiscale features to predict the final change map. (a) Support Correction. (b) Prior-Guided Repair and Decoding. (c) SSF module.

Siamese encoder-decoder architecture based on MobileNetV2 [18]. The two images are fed into a shared encoder to produce four-scale bitemporal features F_a^i and F_b^i , where $i \in \{1, 2, 3, 4\}$ indexes the encoder stages.

Before temporal fusion, RLSC is applied to the encoder features to correct unreliable cross-temporal responses with local support from the opposite temporal branch

$$\{\tilde{F}_a^i, \tilde{F}_b^i\}_{i=1}^4 = \text{RLSC}(\{F_a^i, F_b^i\}_{i=1}^4). \quad (1)$$

The corrected bitemporal features are then fused using shared stagewise fusion (SSF) to obtain a shared multiscale representation

$$F_{sh} = \text{SSF}\left(\left\{\tilde{F}_a^i, \tilde{F}_b^i\right\}_{i=1}^4\right). \quad (2)$$

SSF applies a shared 1×1 projection to each corrected feature pair, fuses the two temporal features by elementwise addition, resizes the four stagewise fused features to the same spatial resolution, and concatenates them along the channel dimension to form F_{sh} .

A lightweight coarse head $\mathcal{H}_c$ predicts the coarse change logit Z_c from F_{sh} , and the corresponding coarse prior is obtained by sigmoid activation

$$Z_c = \mathcal{H}_c(F_{sh})Y_c = \sigma(Z_c) \quad (3)$$

where $\mathcal{H}_c$ consists of a lightweight squeeze residual convolutional block with $1 \times 1 - 3 \times 3 - 1 \times 1$ convolutions, followed by a 1×1 prediction convolution.

Meanwhile, the corrected encoder features are transformed into decoder features $\{L_i\}_{i=1}^4$ through stagewise convolutional blocks. APPR uses the shared representation F_{sh} and the coarse prior Y_c to repair selected low-level decoder features

$$\{\bar{L}_1, \bar{L}_2\} = \text{APPR}(L_1, L_2, F_{sh}, Y_c). \quad (4)$$

APPR takes the probability prior Y_c , while the decoder takes the resized coarse logit Z_c as a pixel-level cue.

The repaired low-level features and the remaining multiscale decoder features are progressively aggregated from deep to shallow stages, and a lightweight prediction head produces the final change logit Z_f . The final map is obtained by

$$\hat{Y} = \sigma(Z_f). \quad (5)$$

B. Reliability-Aware Local Support Correction

RLSC is an encoder-side correction module applied before temporal fusion. To incorporate local evidence beyond corresponding-position temporal comparison, it treats the opposite-temporal neighborhood as a sparse local support set and gates residual correction with support confidence.

As shown in Fig. 3, RLSC consists of sparse local support search and support-checked residual correction. For clarity, we describe how F_a is corrected using local support from F_b . For each spatial position p , RLSC searches a 3×3 local support patch in F_b centered at the corresponding position. The nine positions in this patch are treated as support candidates, indexed by $c = 1, \dots, 9$, and $F_{b,c}(p)$ denotes the feature of the c th support candidate. The opposite direction is computed symmetrically by exchanging F_a and F_b .

1) *Sparse Local Support Search*: As shown in Fig. 3, the routing head $\mathcal{H}_r$ predicts routing scores for local support selection from the current feature, the opposite-temporal feature, and their absolute difference

$$\mathcal{L} = \mathcal{H}_r(F_a, F_b, |F_a - F_b|). \quad (6)$$

At each position p , the routing scores include nine support-candidate entries and additional nonsupport entries. The nonsupport entries represent no-support choices: they participate in softmax normalization but are excluded from feature aggregation.

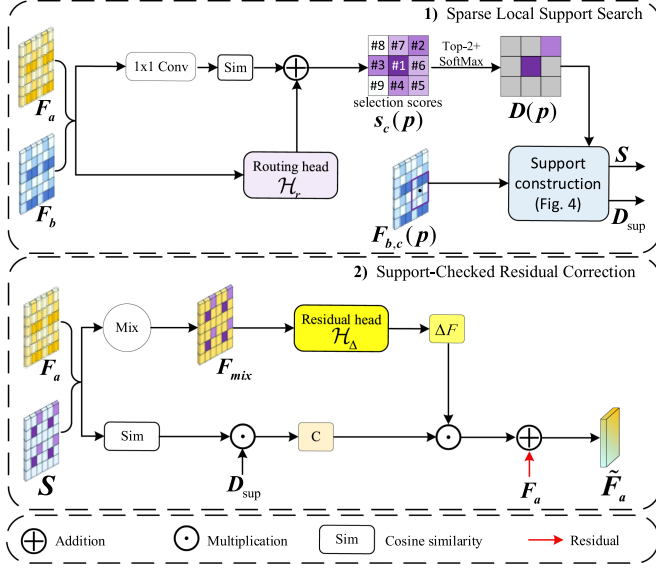


Fig. 3. Illustration of the proposed RLSC.

For the c th support candidate, its selection score is

$$s_c(p) = \text{sim}([\phi(F_a)](p), [\phi(F_b)]_c(p)) + \mathcal{L}_c(p). \quad (7)$$

Here, $\phi(\cdot)$ denotes a lightweight 1×1 projection, and $\text{sim}(\cdot, \cdot)$ denotes channelwise cosine similarity. $\mathcal{L}_c(p)$ denotes the c th support-candidate entry produced by $\mathcal{H}_r$, $s_c(p)$ is used for Top-2 support selection, and $D_c(p)$ denotes the normalized support weight used for aggregation.

RLSC retains the two support candidates with the largest $s_c(p)$ and normalizes their scores together with the nonsupport entries by a softmax. The resulting candidate probabilities form sparse support weights $D(p) = \{D_c(p)\}_{c=1}^9$, while the nonsupport probability is excluded from feature aggregation.

The sparse support weights have two downstream roles. First, their sum gives support availability

$$D_{\text{sup}}(p) = \sum_{c=1}^9 D_c(p), \quad 0 \leq D_{\text{sup}}(p) \leq 1. \quad (8)$$

Since the softmax also includes nonsupport entries, $D_{\text{sup}}(p)$ measures how much routing probability is assigned to local support candidates.

Second, the same sparse support weights are used to aggregate the local support feature

$$S(p) = \sum_{c=1}^9 D_c(p) F_{b,c}(p). \quad (9)$$

Applying this aggregation to all spatial positions yields the support feature map S . Thus, the sparse support weights provide both the support availability map D_{sup} and the support feature map S , as illustrated in Fig. 4.

2) *Support-Checked Residual Correction*: After local support search, RLSC checks whether the support feature S is

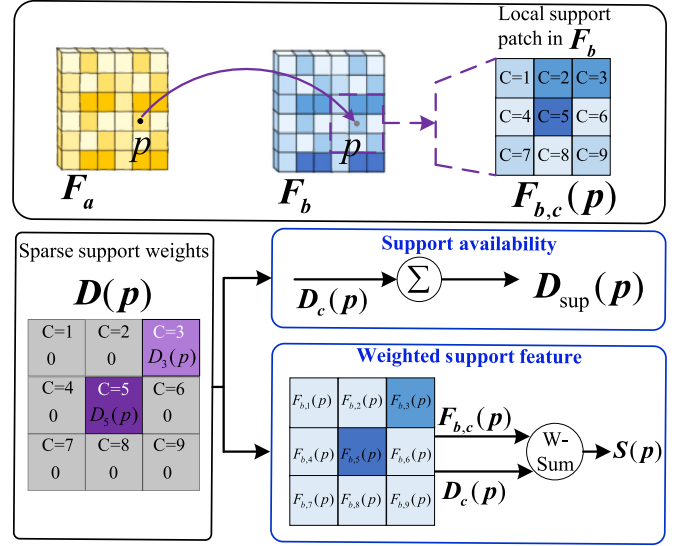


Fig. 4. Construction of support availability and local support feature from sparse support weights. W-Sum denotes support-weighted summation over local candidate features.

consistent with the current feature F_a . The feature-support consistency map is computed as

$$M = \frac{1 + \text{sim}(\phi(F_a), \phi(S))}{2}. \quad (10)$$

The support confidence map is obtained by combining feature-support consistency with support availability

$$C = M \odot D_{\text{sup}}. \quad (11)$$

Thus, C is high only when local support is both available according to routing and consistent according to feature matching.

RLSC then forms a mixed feature from the current feature and the local support feature

$$F_{\text{mix}} = \frac{F_a + S}{2}. \quad (12)$$

The residual head predicts the correction from the current feature, the mixed feature, and their absolute difference. The final update is gated by the support confidence

$$\begin{aligned} \Delta F &= \mathcal{H}_{\Delta}(F_a, F_{\text{mix}}, |F_a - F_{\text{mix}}|) \\ \tilde{F}_a &= F_a + \alpha C \odot \Delta F. \end{aligned} \quad (13)$$

Here, $\mathcal{H}_{\Delta}$ denotes the residual head, and α is a learnable scaling factor. Thus, the residual update is applied mainly where local support is both available and consistent.

For implementation, $\mathcal{H}_r$ uses a compact 1×1 -DW 3×3 - 1×1 head, while $\mathcal{H}_{\Delta}$ uses a compact 1×1 - 3×3 - 1×1 head. Normalization and activation layers are omitted for clarity.

C. Adaptive Prototype-Prior Repair

APPR is designed to repair ambiguous low-level decoder features after the coarse change prior has been produced. The coarse prior provides useful pixel-level semantic guidance in the decoder. However, using the coarse prediction alone to

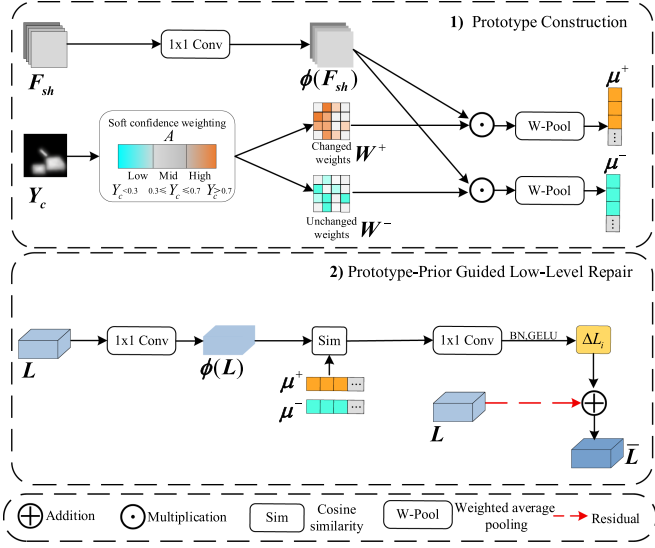


Fig. 5. Illustration of the proposed APPR.

repair low-level features may propagate coarse errors or over-smooth boundaries. Therefore, APPR constructs image-specific changed and unchanged prototypes from the shared representation under confident coarse-prior regions, and uses these prototypes to guide residual repair of selected low-level features. Fig. 5 illustrates the APPR framework.

1) *Prototype Construction*: Given the coarse prior $Y_c = \sigma(Z_c)$ from the coarse head, APPR defines a coarse-prior confidence map

$$A = (2Y_c - 1)^2. \quad (14)$$

As a result, A becomes large in confident changed/unchanged regions and small in uncertain coarse-prior regions. Based on A and Y_c , APPR defines changed and unchanged weights

$$\begin{aligned} W^+ &= A \cdot \text{clip} \left[\frac{Y_c - 0.7}{0.3}, 0, 1 \right] \\ W^- &= A \cdot \text{clip} \left[\frac{0.3 - Y_c}{0.3}, 0, 1 \right] \end{aligned} \quad (15)$$

where $\text{clip}(\cdot, 0, 1)$ denotes an elementwise truncation operator that clips values into $[0, 1]$. The confidence map A suppresses uncertain regions during prototype construction. The weights W^+ and W^- highlight confident changed and unchanged regions, respectively, and are used as spatial weights for prototype pooling rather than as prediction masks. Therefore, only confident coarse-prior regions contribute strongly to prototype construction, while uncertain pixels have limited influence.

Before prototype pooling, the shared representation F_{sh} is projected into a compact prototype space. APPR then extracts one changed prototype and one unchanged prototype by normalized weighted pooling

$$\begin{aligned} \mu^+ &= \text{WPool}(\phi(F_{sh}), W^+) \\ \mu^- &= \text{WPool}(\phi(F_{sh}), W^-). \end{aligned} \quad (16)$$

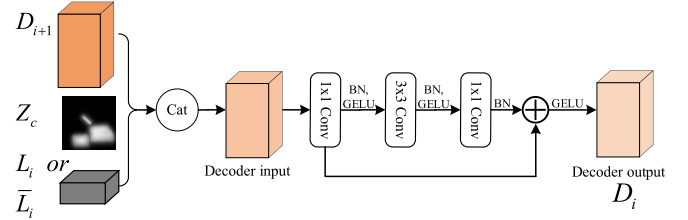


Fig. 6. Illustration of the decoder stage.

Here, $\phi(\cdot)$ denotes a lightweight 1×1 projection, and $\text{WPool}(\cdot)$ denotes weighted average pooling over spatial positions, where the weighted summation is normalized by the sum of weights. The resulting μ^+ and μ^- serve as image-specific changed and unchanged prototypes.

2) *Prototype-Prior-Guided Low-Level Repair*: For a low-level decoder feature L_i , $i \in \{1, 2\}$, we omit the stage index and denote it as L . APPR first projects L into a compact prototype space: $Q = \phi(L)$. At each spatial position, the low-level query is compared with the changed and unchanged prototypes to form a prototype contrast

$$U = \frac{1 + \text{sim}(Q, \mu^+)}{2} \mu^+ - \frac{1 + \text{sim}(Q, \mu^-)}{2} \mu^-. \quad (17)$$

The residual repair signal is predicted from the prototype-guided feature U and the prototype difference $\mu^+ - \mu^-$

$$\Delta L = \text{GELU}(\text{BN}(\text{CONV}_1(\text{Cat}(U, \mu^+ - \mu^-)))) \quad (18)$$

Here, $\text{Cat}(\cdot)$ denotes channelwise concatenation. The prototype difference $\mu^+ - \mu^-$ is broadcast to the size of U before concatenation. The repaired feature is then obtained as

$$\bar{L} = L + \beta \Delta L. \quad (19)$$

Here, β is a learnable factor. The same formulation is applied to L_1 and L_2 with stage-specific parameters.

D. Decoder

After low-level repair, the repaired features and the remaining decoder features are fed into a lightweight hierarchical decoder. The decoder aggregates multiscale features from deep to shallow stages through channelwise concatenation and lightweight residual convolutional blocks. Before concatenation, all decoder inputs are resized to the low-level feature resolution. Fig. 6 illustrates the decoder stage. For APPR-equipped stages, $\bar{L}_i$ is used as the repaired feature; for stages without APPR, we set $\bar{L}_i = L_i$ for notational convenience.

The coarse logit Z_c is resized to the decoder resolution and concatenated with decoder features as a pixel-level coarse cue. The aggregated decoder feature is finally fed into a lightweight 1×1 prediction convolution to produce the final change logit Z_f .

E. Prediction and Loss Function

The final change map is obtained by $\hat{Y} = \sigma(Z_f)$. To supervise both the auxiliary coarse logit and the final logit, we adopt a hybrid change-detection loss composed of binary cross-entropy

TABLE I
KEY CONFIGURATION OF SeCoR

Component	Setting
Backbone	MobileNetV2, ImageNet pretrained
Input size	256×256
Encoder stages	S1: $24 \times 64 \times 64$; S2: $32 \times 32 \times 32$; S3: $96 \times 16 \times 16$; S4: $320 \times 8 \times 8$
RLSC stages	S1-S4
RLSC hidden dim	32
RLSC local support	3×3 window, top-k = 2 from 9 offsets
APPR stages	L1 and L2
APPR hidden dim	16
APPR prototype thresholds	0.7 / 0.3
Decoder channels	64
Prediction	$\sigma(Z_f) > 0.5$, no post-processing
FLOPs protocol	One 256×256 image pair, including encoder, RLSC, SSF, APPR, and decoder

loss and Dice loss. For a prediction logit Z , the loss is defined as

$$\ell_{cd} = \ell_{bce} + \ell_{dice}. \quad (20)$$

Applying this loss to both the auxiliary coarse logit and the final logit gives

$$\mathcal{L} = \ell_{cd}(Z_f, Y^*) + \ell_{cd}(Z_c, Y^*) \quad (21)$$

where Z_f and Z_c denote the final logit and the auxiliary coarse logit, respectively, and Y^* denotes the binary ground-truth change map. During inference, only the final output is used for evaluation $\hat{Y} = \sigma(Z_f)$.

IV. EXPERIMENTS

A. Datasets

Experiments are conducted on four public optical RSCD benchmarks: LEVIR-CD [43], WHU-CD [44], SYSU-CD [45], and UAV-CD [46]. All images are processed into 256×256 inputs. LEVIR-CD follows a 7:1:2 train/validation/test split, WHU-CD follows an 8:1:1 patch split, SYSU-CD follows a 6:2:2 split, and UAV-CD follows an 8:1:1 split.

LEVIR-CD contains 637 pairs of very-high-resolution Google Earth images with a size of 1024×1024 pixels and a spatial resolution of 0.5 m. It mainly focuses on building construction and demolition over long temporal intervals, making it suitable for evaluating building completeness and boundary localization.

WHU-CD consists of one large bitemporal aerial image pair collected in Christchurch, New Zealand, with a size of 32507×15354 pixels and a spatial resolution of 0.2 m. It records post-earthquake reconstruction and contains dense building changes, which are useful for evaluating small buildings, sharp boundaries, and region completeness.

SYSU-CD contains 20 000 pairs of aerial images from Hong Kong, each with a size of 256×256 pixels and a spatial resolution of 0.5 m. It includes diverse urban changes, such as new buildings, suburban expansion, vegetation variation, road expansion, and offshore construction, providing a challenging setting for weak or easily missed changes.

UAV-CD contains 2660 pairs of low-altitude UAV images with a size of 768×768 pixels and a spatial resolution of 0.06 m. It covers building and land changes with fine spatial

details, while viewpoint, illumination, texture, and local geometric variations make it suitable for evaluating pseudochange suppression.

B. Implementation Details

Table I summarizes the key architectural and module settings of SeCoR. SeCoR is implemented in PyTorch and trained on an NVIDIA RTX 4090 GPU with a batch size of 16 and an input size of 256×256 . We use the Adam optimizer with betas of (0.9, 0.99), $\epsilon = 10^{-8}$, and a weight decay of 10^{-4} . The initial learning rate is set to 2.5×10^{-4} and follows a polynomial decay schedule with a power of 0.9, preceded by 200 warm-up iterations.

The training process is controlled by a fixed iteration budget and converted into dataset-specific epochs according to the number of training batches, resulting in 180, 219, 106, and 67 epochs on LEVIR-CD, WHU-CD, SYSU-CD, and UAV-CD, respectively. For each benchmark, all variants and ablations follow the same preprocessing, augmentation, and training pipeline, including normalization, resizing, random crop-resize, random flipping, and temporal image exchange when enabled.

For comparison with state-of-the-art methods, all competing models were reimplemented based on their publicly available code and retrained using the same dataset splits, input resolution, data augmentation strategy, and evaluation protocol. Architecture-specific optimization settings followed the corresponding official implementations, and no additional method-specific hyperparameter tuning was performed. Runtime efficiency was evaluated on an NVIDIA RTX 4090 under FP32 precision with a batch size of 1 and two 256×256 input images. GPU latency was measured using CUDA events after warm-up and averaged over five independent trials. Peak allocated memory was recorded using `torch.cuda.max_memory_allocated()`, excluding data loading and post-processing. The Params and FLOPs in Table III follow the original publications, whereas GPU latency and peak memory were re-measured under this unified runtime protocol.

C. Evaluation Metrics

Following common practice in RSCD, we use precision (Pre.), recall (Rec.), F1-score, and intersection over union (IoU) to

TABLE II
QUANTITATIVE COMPARISON

Method	Source	LEVIR-CD				WHU-CD				SYSU-CD				UAV-CD			
		IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.
FC-EF	ICIP2018	58.66	73.94	70.99	77.16	65.94	79.47	75.32	84.11	54.87	70.86	69.74	72.01	46.15	63.15	57.49	70.06
FC-Siam-Diff	ICIP2018	66.41	79.81	71.43	90.42	49.90	66.58	77.31	58.46	51.20	67.72	58.65	80.12	54.10	70.21	66.70	74.11
FC-Siam-Conc	ICIP2018	73.93	85.01	85.17	84.85	65.98	79.50	74.87	84.75	60.05	75.04	78.38	71.98	44.75	61.83	55.47	69.84
BIT	TGRS2022	80.79	89.37	87.87	90.93	76.60	86.75	89.92	83.79	64.24	78.22	77.34	79.12	54.89	70.87	69.97	71.80
VcT	TGRS2023	81.89	90.04	87.65	92.57	82.61	90.48	90.67	90.29	65.50	79.15	77.29	81.11	54.03	70.16	63.87	77.82
SEIFNet	TGRS2024	83.39	90.94	90.25	91.64	77.70	87.45	89.79	85.23	69.71	82.15	81.12	83.20	57.87	73.31	68.31	79.10
ChangeMamba	TGRS2024	82.09	90.16	88.78	91.59	86.72	92.89	91.99	93.80	71.10	83.11	80.31	86.11	53.90	70.04	66.17	74.40
AEGL-Net	TGRS2025	84.45	91.57	90.85	92.30	88.36	93.82	91.65	96.10	69.70	82.15	78.67	85.95	60.26	75.20	74.17	76.26
CDMamba	TGRS2025	83.68	91.12	90.50	91.74	85.88	92.40	91.81	93.00	67.14	80.34	79.41	81.29	56.84	72.48	68.04	77.54
TMSF	JSTARS2025	83.06	90.74	89.33	92.20	82.01	90.12	87.49	92.91	66.37	79.78	74.55	85.82	56.43	72.15	69.14	75.44
B2CNet	JSTARS2024	83.65	91.10	91.02	91.18	87.04	93.07	91.79	94.39	69.18	81.78	83.09	80.51	55.10	71.06	68.22	74.13
LCD-Net	JSTARS2025	84.31	91.49	89.67	93.38	86.88	92.98	91.79	94.20	68.36	81.21	79.33	83.18	60.65	75.50	74.04	77.03
ChangeFormer	IGARSS2022	82.98	90.70	89.78	91.64	84.05	91.33	88.99	93.80	65.00	78.79	77.86	79.74	47.38	64.30	61.29	67.62
STNet-ResNet50	ICME 2023	82.38	90.34	92.00	88.73	84.24	91.45	92.74	90.19	70.75	82.87	84.61	81.21	54.46	70.51	74.66	66.80
RFANet	ISPRS2024	84.67	91.70	90.76	92.66	89.22	94.30	92.36	96.32	71.22	83.19	80.80	85.73	60.53	75.41	74.72	76.11
SARAS-Net	AAAI2023	81.44	89.77	88.24	91.36	82.57	90.45	91.34	89.58	65.62	79.24	73.21	86.35	51.80	68.25	68.07	68.42
Ours	-	84.74	91.74	90.83	92.67	89.98	94.72	93.52	95.96	72.52	84.07	82.53	85.68	63.53	77.69	78.82	76.61

Competing methods are reported using reproduced single-run results under the stated evaluation protocol, whereas the SeCoR entry is the mean over five random-seed runs. Bold values indicate the highest reported scores.

TABLE III
COMPARISON OF MODEL ARCHITECTURES AND EFFICIENCY

Method	Params (M)	FLOPs (G)	GPU Latency (ms/pair)	Peak Mem. (MiB)
FC-EF	1.35	3.57	2.51	56.32
FC-Siam-Diff	1.35	4.72	3.76	67.19
FC-Siam-Conc	1.55	5.33	3.50	70.88
B2CNet	16.10	22.36	14.63	203.30
SEIFNet	27.91	8.37	9.91	153.39
SARAS-Net	32.33	60.64	13.44	680.54
BIT	3.50	10.63	11.55	84.08
VcT	3.57	10.64	35.12	314.59
TMSF	12.32	11.80	36.41	95.13
ChangeFormer	41.03	202.79	17.52	461.16
STNet-ResNet50	48.77	44.97	13.26	418.93
ChangeMamba	49.94	114.82	40.64	392.08
CDMamba	11.91	54.18	51.62	235.30
LCD-Net	2.56	4.45	56.04	54.59
RFANet	2.86	3.16	24.68	71.31
AEGL-Net	3.85	1.26	22.43	60.84
SeCoR (Ours)	2.50	2.66	26.76	46.70

evaluate detection performance. These metrics are defined as

$$\text{Pre.} = \frac{TP}{TP + FP} \quad \text{Rec.} = \frac{TP}{TP + FN} \quad (22)$$

$$F1 = \frac{2 \cdot \text{Pre.} \cdot \text{Rec.}}{\text{Pre.} + \text{Rec.}} \quad \text{IoU} = \frac{TP}{TP + FP + FN} \quad (23)$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. All accuracy values in the tables are reported as percentages, and the same counting protocol is used for all SeCoR variants and ablation settings. The controlled ablation, diagnostic-subset, and backbone analyses in Table IV–Table IX use the same prespecified reference seed for all compared variants. Table X reports five-run mean $\pm$ SD and the seed-matched statistical comparison between SeCoR and the architecture-matched Baseline.

D. Comparison With State-of-the-Art Methods

1) *Quantitative and Efficiency Comparison:* Table II compares SeCoR with 16 representative RSCD methods on four benchmarks. The comparison includes early fully convolutional and Siamese baselines, namely FC-EF, FC-Siam-Diff, and FC-Siam-Conc [5]; CNN-based, lightweight, or boundary detail-aware methods, including B2CNet [14], SEIFNet [17], AEGL-Net [48], LCD-Net [38], and RFANet [39]; Transformer or hybrid spatiotemporal models, including

BIT [8], VcT [24], TMSF [40], ChangeFormer [47], STNet-ResNet50 [25], and SARAS-Net [26]; and Mamba-based methods, including ChangeMamba [28] and CDMamba [29]. To ensure a fair comparison, we reimplemented these methods based on their open-source code. Table III summarizes their model architectures, computational costs, and runtime efficiency.

First, SeCoR achieves the highest F1 and IoU on all four datasets, with marginal gains on LEVIR-CD, a modest gain on WHU-CD, and larger gains on SYSU-CD and UAV-CD. These results indicate that SeCoR maintains competitive performance on building-dominated benchmarks while providing larger empirical gains in scenes involving weak changes, cluttered backgrounds, viewpoint variation, and appearance-induced pseudochanges.

Second, SeCoR remains compact, with 2.50M parameters and 2.66G FLOPs. Compared with heavier attention-, Transformer-, or Mamba-based methods, it requires lower model complexity while achieving comparable or better accuracy. SeCoR also requires 26.76 ms per image pair and 46.70 MiB of peak GPU memory. Although its latency is not the lowest, it achieves the lowest peak memory among all compared methods, providing a favorable balance between detection accuracy and deployment cost.

2) *Qualitative Comparison:* The qualitative results further explain the quantitative trends in Table II. Instead of only showing average accuracy, Figs. 7–10 illustrate typical failure patterns, including incomplete changed objects, boundary fringes,

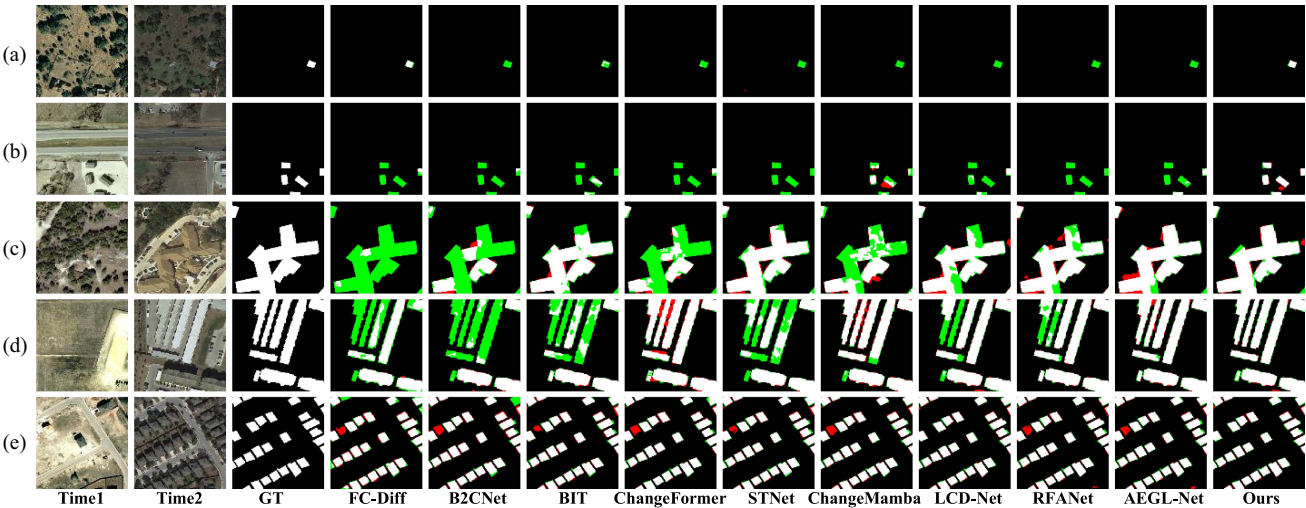


Fig. 7. Visualization results of methods on LEVIR-CD. (a)–(e) Representative samples selected from the LEVIR-CD dataset.

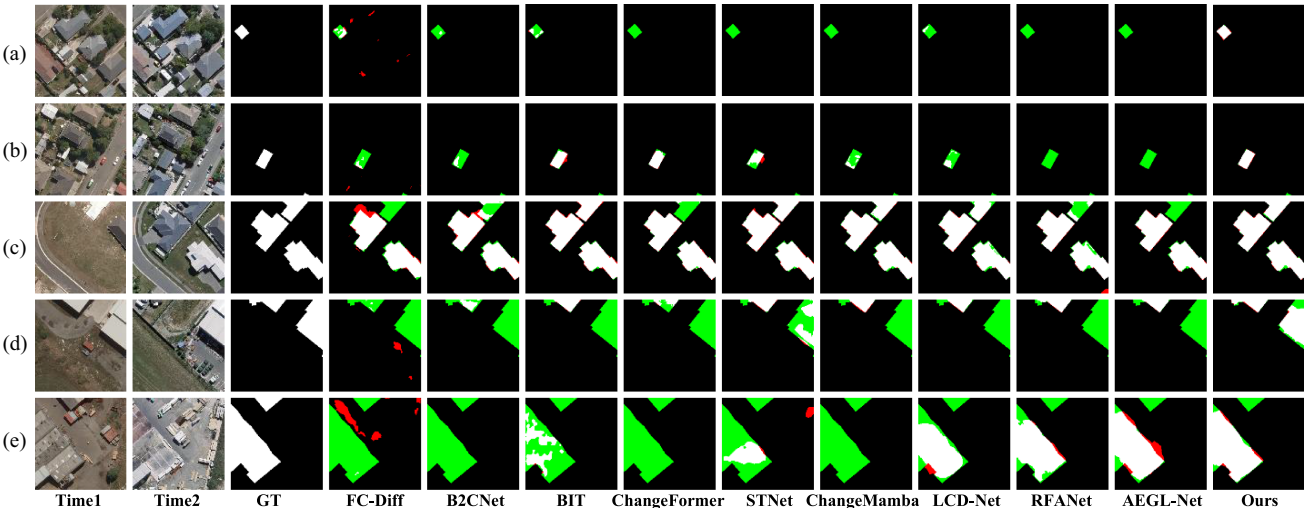


Fig. 8. Visualization results of methods on WHU-CD. (a)–(e) Representative samples selected from the WHU-CD dataset.

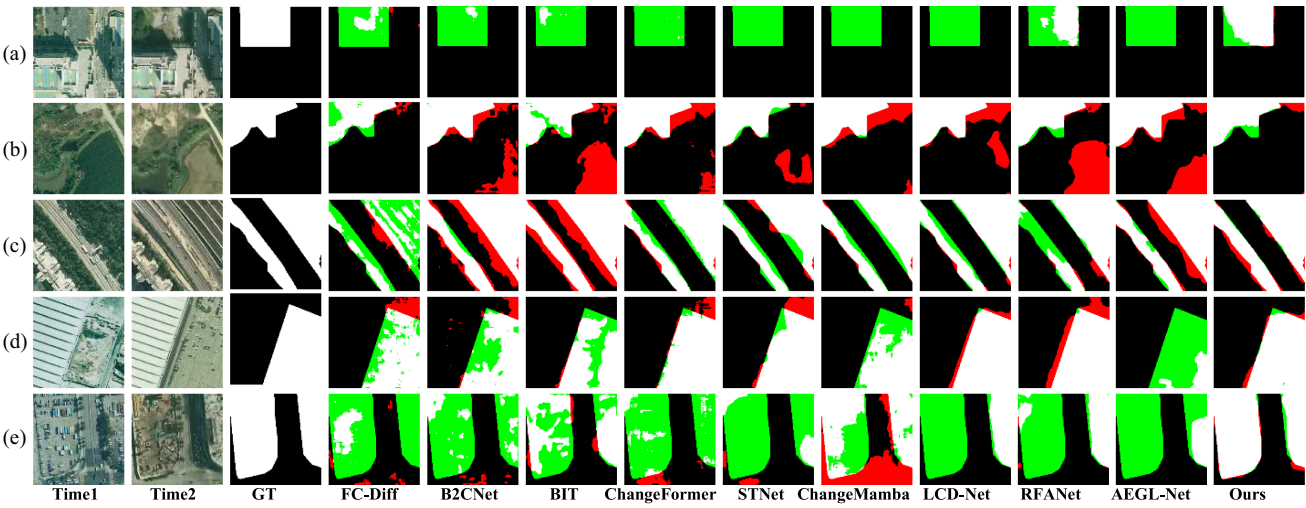


Fig. 9. Visualization results of methods on SYSU-CD. (a)–(e) Representative samples selected from the SYSU-CD dataset.

TABLE IV
SYSTEM ABLATION OF RLSC AND APPR

Method	LEVIR-CD				WHU-CD				SYSU-CD				UAV-CD				Params (M)	FLOPs (G)
	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.		
Baseline	81.68	89.91	89.20	90.65	85.86	92.39	90.83	94.01	66.75	80.06	75.69	84.96	58.38	73.72	76.01	71.57	2.28	2.24
Baseline+RLSC	83.97	91.29	90.35	92.24	89.14	94.26	93.23	95.32	71.25	83.21	82.23	84.22	62.18	76.68	77.96	75.43	2.45	2.44
Baseline+APPR	84.05	91.33	90.88	91.79	88.76	94.04	91.63	96.58	70.22	82.51	80.48	84.64	60.78	75.60	78.94	72.54	2.34	2.46
Full	84.72	91.73	90.60	92.89	90.19	94.84	93.48	96.24	72.58	84.11	81.66	86.71	63.91	77.98	78.28	77.68	2.50	2.66

TABLE V
COMPARISON OF TEMPORAL INTERACTION DESIGNS

Method	LEVIR-CD				WHU-CD				SYSU-CD				UAV-CD				Params (M)	FLOPs (G)
	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.		
Sub	83.12	90.78	89.50	92.10	89.18	94.28	93.17	95.42	71.00	83.04	79.38	87.06	62.48	76.90	74.16	79.86	2.41	2.47
Cat	84.56	91.64	90.72	92.57	87.89	93.56	92.30	94.85	69.51	82.02	78.22	86.20	61.50	76.16	80.40	72.35	2.44	2.52
SPG	84.41	91.55	90.66	92.45	89.24	94.32	92.59	96.11	70.86	82.95	80.56	85.48	61.70	76.31	77.02	75.62	2.49	2.61
LK	84.60	91.66	90.72	92.61	89.62	94.53	93.53	95.54	71.25	83.21	79.47	87.31	62.41	76.85	82.30	72.08	3.87	4.37
RLSC	84.72	91.73	90.60	92.89	90.19	94.84	93.48	96.24	72.58	84.11	81.66	86.71	63.91	77.98	78.28	77.68	2.50	2.66

Best results are highlighted in bold values.

TABLE VI
SUPPORT TOP-K SENSITIVITY OF RLSC

Support Top-k	LEVIR-CD				WHU-CD				SYSU-CD				UAV-CD				Params (M)	FLOPs (G)
	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.		
Top-k = 1	84.52	91.61	90.76	92.48	89.32	94.36	91.86	97.00	71.93	83.67	80.61	86.98	63.22	77.46	83.04	72.59	2.41	2.47
Top-k = 2	84.72	91.73	90.60	92.89	90.19	94.84	93.48	96.24	72.58	84.11	81.66	86.71	63.91	77.98	78.28	77.68	2.50	2.66
Top-k = 4	84.41	91.55	91.01	92.09	89.48	94.45	92.86	96.09	72.15	83.82	82.56	85.13	62.76	77.12	82.96	72.05	2.49	2.61
Top-k = 9	84.36	91.52	91.02	92.02	88.99	94.18	93.09	95.28	71.96	83.69	81.11	86.44	62.58	76.98	82.09	72.47	2.44	2.52

TABLE VII
ACTIVE STAGE CONFIGURATION FOR APPR

Method	Repaired Stages	LEVIR-CD				WHU-CD				SYSU-CD				UAV-CD				Params (M)	FLOPs (G)
		IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.		
APPR L1 Only	L1	84.54	91.62	90.88	92.38	89.68	94.55	93.21	95.94	71.98	83.71	83.58	83.84	62.92	77.24	74.95	79.68	2.41	2.47
APPR L2 Only	L2	84.16	91.40	90.40	92.42	89.29	94.34	93.04	95.68	71.28	83.23	79.85	86.92	62.37	76.82	77.15	76.50	2.44	2.52
APPR L1+L2+L3+L4	L1-L4	84.49	91.59	91.00	92.19	89.39	94.40	93.16	95.68	71.91	83.66	80.84	86.67	62.81	77.16	74.02	80.58	2.49	2.61
APPR	L1,L2	84.72	91.73	90.60	92.89	90.19	94.84	93.48	96.24	72.58	84.11	81.66	86.71	63.91	77.98	78.28	77.68	2.50	2.66

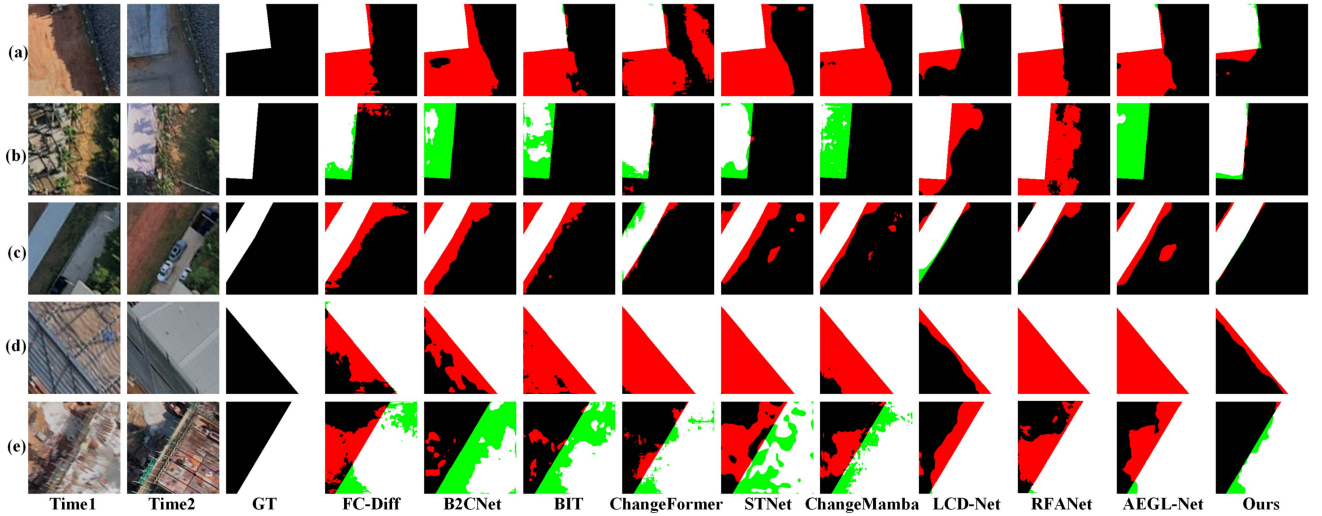


Fig. 10. Visualization results of methods on UAV-CD. (a)–(e) Representative samples selected from the UAV-CD dataset.

weak changed regions, and pseudochange responses caused by appearance variation. Here, white, black, red, and green denote true positives, true negatives, false positives, and false negatives, respectively.

In these examples, the main differences are false positives in unchanged background regions and false negatives inside

changed objects. The following datasetwise discussion describes the observed failure types.

Fig. 7 focuses on LEVIR-CD, where the selected samples mainly contain building-related changes. Rows (a) and (b) contain small or weakly contrasted changed targets, including narrow structures and building-related changes. Rows (c), (d),

TABLE VIII
FULL-TEST AND DIAGNOSTIC-SUBSET RESULTS ON SYSU-CD AND UAV-CD

Method	SYSU-CD				UAV-CD			
	IoU	F1	Rec.	Pre.	IoU	F1	Rec.	Pre.
Baseline	66.75	80.06	75.69	84.96	58.38	73.72	76.01	71.57
Baseline (subset)	58.24	73.61	68.19	79.98	55.87	71.69	81.15	64.20
SeCoR (subset)	68.37	81.21	79.68	82.81	63.20	77.45	84.13	71.75
SeCoR	72.58	84.11	81.66	86.71	63.91	77.98	78.28	77.68

TABLE IX
ABLATION STUDY WITH DIFFERENT BACKBONES ON THE LEVIR-CD, WHU-CD, SYSU-CD, AND UAV-CD DATASETS

Method	LEVIR-CD		WHU-CD		SYSU-CD		UAV-CD		Params (M)	FLOPs (G)
	IoU	F1	IoU	F1	IoU	F1	IoU	F1		
ResNet18	83.86	91.22	87.48	93.32	67.81	80.82	58.74	74.01	12.41	7.17
ResNet50	84.00	91.30	89.54	94.48	69.15	81.76	59.69	74.76	31.39	17.24
MobileNetV2(0.5x)	83.35	90.92	88.55	93.93	70.27	82.54	60.78	75.61	0.99	1.99
SeCoR	84.72	91.73	90.19	94.84	72.58	84.11	63.91	77.98	2.50	2.66

TABLE X
FIVE-RUN STABILITY AND SEED-LEVEL PAIRED TESTS AGAINST THE ARCHITECTURE-MATCHED BASELINE

Dataset	Baseline (mean $\pm$ SD)		SeCoR (mean $\pm$ SD)		Gain (pp)		Paired t-test pHolm (F1)
	IoU	F1	IoU	F1	Δ IoU	Δ F1	
LEVIR-CD	81.70 $\pm$ 0.10	89.93 $\pm$ 0.06	84.74 $\pm$ 0.05	91.74 $\pm$ 0.03	+3.04	+1.81	<0.001
WHU-CD	85.86 $\pm$ 0.19	92.39 $\pm$ 0.11	89.98 $\pm$ 0.14	94.72 $\pm$ 0.08	+4.12	+2.33	<0.001
SYSU-CD	66.75 $\pm$ 0.22	80.06 $\pm$ 0.16	72.52 $\pm$ 0.18	84.07 $\pm$ 0.12	+5.77	+4.01	<0.001
UAV-CD	58.38 $\pm$ 0.28	73.72 $\pm$ 0.23	63.53 $\pm$ 0.23	77.69 $\pm$ 0.17	+5.15	+3.97	<0.001

F1 and IoU are reported as percentages in the form mean $\pm$ SD over five random-seed runs, where SD denotes the sample standard deviation. The two-sided paired t-test uses the five seed-matched global F1 scores of the architecture-matched Baseline and SeCoR ($df = 4$). Because F1 and IoU are derived from the same global confusion counts, statistical inference is conducted for F1 only. The resulting p-values are Holm-adjusted across the four datasets.

and (e) involve elongated buildings, dense residential areas, and large building regions with narrow boundaries. In these cases, many methods produce red-green boundary fringes, indicating inaccurate localization along roof edges and adjacent structures. The proposed method preserves more complete building extents and reduces peripheral false alarms in these samples.

Fig. 8 presents results on WHU-CD, where the visual comparison mainly reflects high-resolution building geometry. Rows (a) and (b) contain small and isolated building changes, where several competing methods miss parts of the targets or introduce scattered false alarms around visually similar structures. Rows (c), (d), and (e) involve larger buildings and adjacent structural boundaries, for which inaccurate methods often show boundary expansion, broken interiors, or edge leakage. The proposed method produces fewer fragmented interiors and scattered false responses, although missed interiors remain in some large oblique building regions.

Fig. 9 illustrates SYSU-CD scenes with illumination variation, weak object contrast, shoreline-like structures, and large-area background changes. Rows (a) and (b) contain broad changed regions with ambiguous boundaries, where several methods over-activate surrounding background areas and generate large false-positive regions. Rows (c) and (d) include oblique boundaries along roads, water bodies, or land-cover transitions, causing many competing methods to produce continuous boundary errors. Row (e) contains extensive changed regions on both sides of a road corridor, with several methods producing large false-negative areas. In these samples, the proposed method better constrains overactivation and preserves more coherent main changed regions.

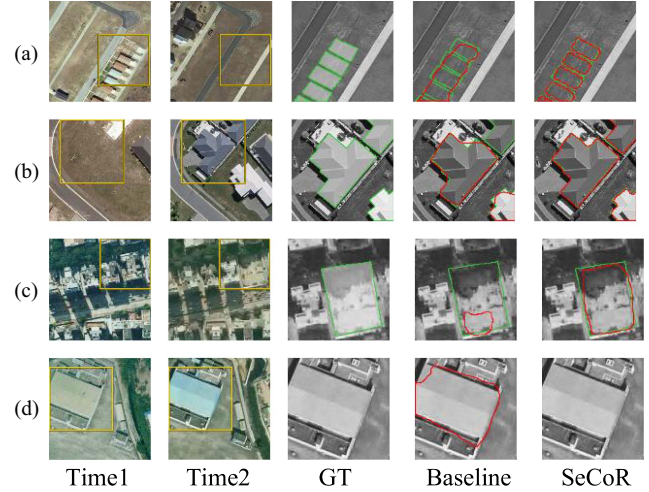


Fig. 11. Enlarged comparison of representative boundary-ambiguity, missed-change, and pseudochange cases. Yellow boxes locate the enlarged regions; green and red contours denote ground-truth and predicted boundaries, respectively. No green contour in (d) indicates an unchanged region. (a) Boundary ambiguity involving adjacent small structures. (b) Boundary ambiguity involving a large irregular footprint. (c) Missed-change (under-response) case. (d) Pseudo-change (over-response) case.

Fig. 10 shows UAV-CD examples that highlight the difficulty of low altitude change detection, where viewpoint shifts, scale variation, and texture changes often lead to pseudochanges. In rows (a), (b), and (c), several competing methods incorrectly respond to road edges, bare soil, and locally shifted textures, resulting in extensive false-positive regions. Rows (d) and (e) include large oblique regions and irregular boundaries, where some methods either miss substantial changed areas or produce fragmented masks. The proposed method suppresses many irrelevant background responses and preserves more compact shapes for the actual changed regions, although slight boundary deviations remain in some large oblique areas.

Overall, the visual comparisons reveal two major error types. On LEVIR-CD and WHU-CD, the differences mainly appear around building interiors, fragmented small objects, and roof boundaries, where APPR helps preserve low-level spatial details. On SYSU-CD and UAV-CD, errors are more related to weak visual evidence and pseudochanges caused by illumination, viewpoint, seasonal, or texture variations, where RLSC helps suppress unreliable cross-temporal responses, consistent with the quantitative improvements in Table II.

To clarify the local behavior behind the aggregate results in Figs. 7–10, Fig. 11 presents enlarged comparisons between the baseline and SeCoR in representative challenging regions. In Fig. 11(a), the baseline produces rounded and partially connected contours for adjacent small structures, whereas SeCoR better preserves the separation between neighboring structures and their rectilinear geometry. In Fig. 11(b), SeCoR more closely follows the extent and corner transitions of a large irregular footprint, reducing the local contraction and boundary displacement observed in the baseline. These cases indicate finer spatial agreement rather than a simple expansion of the predicted change regions, consistent with the subsequent BF1 analysis.

Fig. 11(c) represents an under-response case: the baseline activates only a small portion of a spatially coherent changed region,

whereas SeCoR restores most of its annotated extent. Fig. 11(d) represents an over-response case: although the highlighted roof exhibits substantial cross-temporal appearance variation, the region is unchanged according to the annotation. The baseline follows this apparent difference and produces a pseudochange response, whereas SeCoR suppresses it. The opposite correction directions suggest that SeCoR reshapes the response distribution conditionally rather than uniformly increasing or decreasing change activation.

These visual results complement quantitative comparison. On LEVIR-CD and WHU-CD, the gains mainly appear as fewer boundary fringes and less fragmented building regions. On SYSU-CD, they are more evident in recovering missed changes in weak or cluttered scenes. On UAV-CD, they mainly reflect better suppression of pseudochange responses caused by appearance and viewpoint variations.

E. Ablation Studies

The ablation studies examine the contribution of the proposed correction modules and the robustness of the main design choices. The baseline denotes the same lightweight Siamese encoder-decoder architecture without RLSC and APPR. All variants use the same backbone, decoder, preprocessing, training schedule, and inference setting as the full model.

1) *Effect of RLSC and APPR*: Table IV shows that both RLSC and APPR consistently improve the baseline on all four datasets. RLSC gives larger gains on SYSU-CD and UAV-CD, suggesting that reliability-checked local support is beneficial in scenes with weak changes and pseudochange disturbance. APPR also improves the Baseline across all four datasets, showing that decoder-side prototype-prior repair provides an additional contribution. The Full model achieves the best F1 and IoU on all four datasets, supporting the complementarity of the two correction stages.

Fig. 12 complements the module ablation with heatmap visualizations of the baseline, the two single-module variants, and the full model. Warmer colors indicate stronger change responses. On LEVIR-CD and WHU-CD, the single-module variants and the full model show more concentrated responses around building regions and fewer scattered high-response areas near adjacent structures. On SYSU-CD, the full model gives stronger responses in several weak true-change regions. On UAV-CD, the correction modules reduce diffuse responses in some unchanged areas while preserving the main changed regions. These visual results suggest that the two modules reshape the response distribution rather than simply increasing activation strength everywhere.

2) *Comparison of Temporal Interaction Designs*: Table V compares RLSC with feature subtraction, channel concatenation, same-pixel gated fusion (SPG), and a larger-kernel convolutional fusion block (LK) under the same backbone, decoder, coarse cue, and training protocol. RLSC achieves the best F1 and IoU on all four datasets with a computational budget close to simple fusion variants and much lower than large-kernel fusion. These results suggest that the advantage of RLSC comes from reliability-aware support selection rather than merely stronger local fusion capacity.

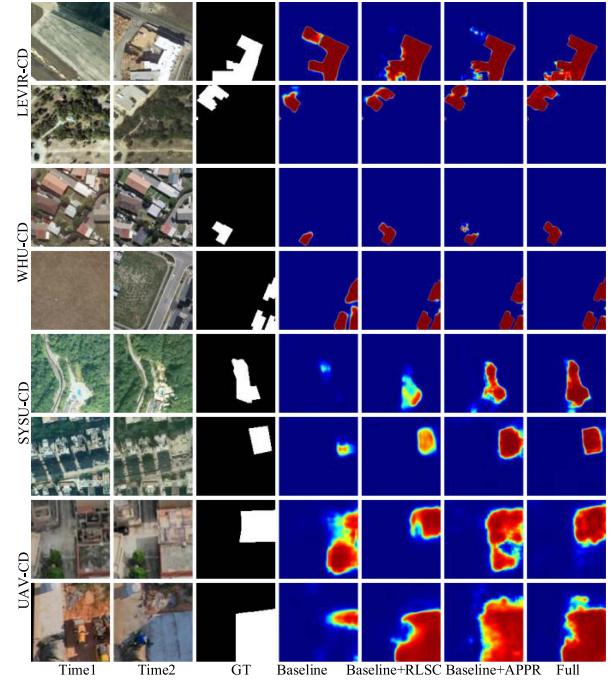


Fig. 12. Heatmap visualization of the baseline, single-module settings, and the full model.

On SYSU-CD, it reaches 84.11% F1 and 72.58% IoU, outperforming LK while using fewer parameters and FLOPs. On UAV-CD, it obtains 77.98% F1 and 63.91% IoU, while all alternatives yield lower F1 and IoU under strong appearance disturbance.

These results indicate that the improvement is not simply due to increased fusion capacity. Instead, RLSC benefits from selecting reliable local cross-temporal support before feature correction, making it a selective correction mechanism rather than dense local attention.

3) *Support Top-k Sensitivity*: Table VI studies the number of support candidates retained in the local routing step. The default setting uses $k = 2$. When $k = 1$, the routing is highly sparse and can avoid noisy support aggregation, but it may discard valid support when multiple nearby positions provide useful cross-temporal evidence. Using $k = 2$ achieves the best F1 on LEVIR-CD, WHU-CD, SYSU-CD, and UAV-CD, showing a better balance between selectivity and support coverage.

These results indicate that retaining more local candidates does not necessarily improve RLSC. A compact support set provides a better balance between support coverage and noisy-match suppression, supporting the fixed Top-2 setting used in RLSC.

4) *Active Decoder Stages for APPR*: Table VII studies, which decoder stages should receive prototype-prior repair. Applying the module only to L_1 performs strongly on building-dominated datasets, where fine spatial details are important for boundary recovery. Using only L_2 is less effective on SYSU-CD, suggesting that shallow contextual information alone is insufficient for weak changed regions. The default $L_1 + L_2$ setting achieves the best overall F1 and IoU, while extending the module to all decoder stages brings no further gain. This indicates that prototype-prior

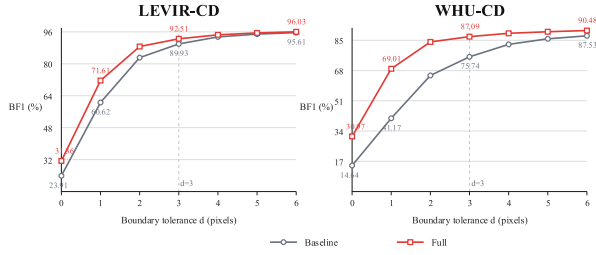


Fig. 13. Boundary F1 curves on LEVIR-CD and WHU-CD.

repair is most suitable for low-level decoder features rather than generic all-stage enhancement.

Extending APPR to all decoder stages does not further improve F1 or IoU. This suggests that APPR is more suitable for low-level detail repair than generic all-stage enhancement, since higher decoder stages already contain stronger semantic context.

F. Boundary Accuracy Analysis

Following the boundary-matching evaluation protocol used in boundary detection [58], we evaluate boundary accuracy using Boundary F1 (BF1) with a spatial tolerance d . Given the predicted and ground-truth change boundaries, boundary precision and boundary recall are computed by matching boundary pixels within d pixels, and BF1 is defined as

$$\text{BF1}(d) = \frac{2\text{TP}_b}{2\text{TP}_b + \text{FP}_b + \text{FN}_b} \quad (24)$$

where TP_b , FP_b , and FN_b denote matched boundary pixels, unmatched predicted boundary pixels, and unmatched ground-truth boundary pixels under tolerance d , respectively. We report BF1 on LEVIR-CD and WHU-CD because both benchmarks are dominated by building changes and provide relatively clear object-level change boundaries.

Fig. 13 shows that the full model remains above the baseline across different tolerances for $d = 0, \dots, 6$. The advantage is more evident under stricter matching. Compared with the baseline, BF1@1 improves by 10.99 and 27.84 percentage points on LEVIR-CD and WHU-CD, respectively. At BF1@3, the gains remain positive, with improvements of 2.58 and 11.35 percentage points on LEVIR-CD and WHU-CD, respectively.

A smaller d imposes a stricter requirement on boundary localization, whereas a larger d accepts moderate spatial deviations. As the tolerance becomes larger, the gap gradually narrows because coarse boundary predictions are more easily matched.

Therefore, the larger gains at small tolerances indicate improved fine boundary agreement rather than only changed-region expansion. This analysis supports low-level repair and is consistent with the APPR-stage ablation.

G. Diagnostic Subset Results

SYSU-CD and UAV-CD exhibit two opposite forms of unreliable visual evidence. In SYSU-CD, weak bitemporal

contrast in true changed regions often causes missed detections. In UAV-CD, viewpoint, illumination, texture, and local geometric variations can produce strong apparent differences in unchanged regions, leading to false alarms. Recent hard-region analyses likewise identify change boundaries and background-induced pseudochanges as persistent challenges in RSCD [13], [59]. We, therefore, construct two prediction-independent diagnostic subsets using only T_1 , T_2 , and Y^* .

For each test pair, we compute a pixelwise appearance-difference score $E(p) = ||T_1(p) - T_2(p)||_1$ after the same normalization as model input. T_1 and T_2 provide the visual difference, while Y^* only determines the averaging region.

For SYSU-CD, samples are ranked by the average $E(p)$ over the ground-truth changed region, and the lowest 25% of test samples are selected as low-evidence true-change cases.

For UAV-CD, samples with a small changed-area ratio are first retained according to Y^* , then ranked by the average $E(p)$ over the unchanged region, and the highest 25% of test samples are selected as near-unchanged disturbance cases. All subsets are fixed before model evaluation.

Table VIII shows improvements on both diagnostic subsets. On SYSU-CD, the largest gain is in Recall (+11.49 percentage points), consistent with better recovery of weak true changes. On UAV-CD, the largest gain is in Precision (+7.55 percentage points), consistent with fewer appearance-induced false alarms. F1 increases by 7.60 and 5.76 percentage points on the two subsets, respectively.

H. Backbone Sensitivity

To examine whether the proposed correction design depends on a specific encoder capacity, we replace the default MobileNetV2 backbone with ResNet-18 and ResNet-50 [9] while keeping RLSC, APPR, the decoder, input size, and training protocol unchanged. We also include a width-reduced MobileNetV2(0.5x) variant, by reducing the backbone channel width to evaluate the compressibility of SeCoR. Table IX reports F1/IoU together with parameter number.

The results show that increasing backbone capacity does not consistently improve change detection accuracy. The ResNet-based variants remain competitive, but their additional parameters do not yield a better accuracy-efficiency tradeoff. Reducing MobileNetV2 from $1.0\times$ to $0.5\times$ decreases the parameter count from 2.50M to 0.99M and FLOPs from 2.66G to 1.99G. The more moderate FLOP reduction arises because several decoder and correction components remain fixed-width. SeCoR retains competitive accuracy under the reduced backbone budget; therefore, MobileNetV2 ($1.0\times$) is adopted for the best overall balance, while the $0.5\times$ variant provides a compact alternative under tighter parameter budgets.

V. DISCUSSION

Table X summarizes five seed-matched runs of the architecture-matched Baseline and SeCoR under identical settings. SeCoR achieves higher mean F1 and IoU on all four

datasets, with mean F1 gains substantially exceeding the corresponding run-to-run standard deviations. All four seed-level paired t-tests on F1 remain significant after Holm adjustment (adjusted $p < 0.001$).

F1 and IoU are reported as percentages in the form mean $\pm$ SD over five random-seed runs, where SD denotes the sample standard deviation. The two-sided paired t-test uses the five seed-matched global F1 scores of the architecture-matched Baseline and SeCoR ($df = 4$). Because F1 and IoU are derived from the same global confusion counts, statistical inference is conducted for F1 only. The resulting p-values are Holm-adjusted across the four datasets.

The qualitative examples in Fig. 11 illustrate representative residual errors around building boundaries and fragmented structures, together with cases involving weak true changes and appearance-induced pseudochanges. Fig. 13 further shows that the full model remains above the Baseline across the evaluated boundary tolerances, with a larger advantage under stricter matching. The diagnostic results in Table VIII likewise show higher full-model F1 and IoU than the Baseline on the predefined low-evidence true-change and near-unchanged disturbance subsets. These qualitative, boundary, and diagnostic results are interpreted at the full-model level.

The module-level ablations in Table IV show F1 and IoU gains from adding either RLSC or APPR to the architecture-matched Baseline on all four datasets, while their combination achieves the highest values. These results support complementary contributions at the aggregate dataset level. Mechanistically, the two modules implement distinct correction pathways: RLSC performs support-checked local correction before temporal fusion, whereas APPR performs prototype-guided repair at two designated low-level decoder stages after coarse prediction, using image-specific prototypes constructed from confident coarse-prior regions.

From a deployment perspective, RLSC relies on sparse local support rather than dense cross-temporal matching, while APPR is applied only at two designated low-level decoder stages. Table III shows that the complete SeCoR remains compact, requiring only 2.50 M parameters and 2.66G FLOPs, with a latency of 26.76 ms per image pair and a peak GPU memory usage of 46.70 MiB under the stated evaluation setting. These results suggest potential for resource-constrained deployment; however, the hardware-specific measurements do not establish real-time performance on edge devices. Several limitations also remain. Under severe geometric displacement or large viewpoint changes, RLSC can attenuate unsupported updates but cannot recover cross-temporal evidence outside its local search region. APPR may be affected when confidently incorrect coarse predictions bias prototype estimation. In addition, the current evaluation does not establish cross-domain transfer, large-area operational performance, or real-time edge inference.

VI. CONCLUSION

In this study, we presented SeCoR, an evidence-guided selective correction framework for lightweight RSCD. SeCoR combines support-checked encoder correction with prototype-

prior decoder repair: RLSC corrects encoder features before temporal fusion using reliability-checked local support, while APPR uses the shared representation and confident coarse-prior regions to construct image-specific prototype references for low-level feature repair. Experiments on LEVIR-CD, WHU-CD, SYSU-CD, and UAV-CD demonstrate a competitive accuracy-efficiency tradeoff with only 2.50 M parameters and 2.66 G FLOPs, while the half-width variant remains effective under a stricter lightweight budget. The five-run analysis using five different random seeds further confirms stable and reproducible improvements over the architecture-matched Baseline across all four benchmarks. Ablation, boundary, diagnostic-subset, and backbone analyses support the complementary roles of RLSC and APPR. Together with the measured runtime and memory footprint, these results indicate practical potential for resource-constrained deployment. Future work will investigate adaptive support selection, cross-domain generalization, large-area operational deployment, and hardware-oriented optimization for edge devices.

REFERENCES

- [1] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogrammetry Remote Sens.*, vol. 80, pp. 91–106, 2013.
- [2] R. E. Kennedy et al., "Remote sensing change detection tools for natural resource managers: Understanding concepts and tradeoffs in the design of landscape monitoring projects," *Remote Sens. Environ.*, vol. 113, no. 7, pp. 1382–1396, 2009.
- [3] G. Cheng et al., "Change detection methods for remote sensing in the last decade: A comprehensive review," *Remote Sens.*, vol. 16, no. 13, Jun. 2024, Art. no. 2355.
- [4] D. Peng, X. Liu, Y. Zhang, H. Guan, Y. Li, and L. Bruzzone, "Deep learning change detection techniques for optical remote sensing imagery: Status, perspectives and challenges," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 136, Feb. 2025, Art. no. 104282.
- [5] R. C. Daudt, B. L. Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process.*, Oct. 2018, pp. 4063–4067.
- [6] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogrammetry Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.
- [7] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
- [8] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.
- [9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
- [10] A. P. Tewkesbury, A. J. Comber, N. J. Tate, A. Lamb, and P. F. Fisher, "A critical synthesis of remotely sensed optical image change detection techniques," *Remote Sens. Environ.*, vol. 160, pp. 1–14, Apr. 2015, doi: [10.1016/j.rse.2015.01.006](https://doi.org/10.1016/j.rse.2015.01.006).
- [11] R. J. Radke, S. Andra, O. Al-Kofahi, and B. Roysam, "Image change detection algorithms: A systematic survey," *IEEE Trans. Image Process.*, vol. 14, no. 3, pp. 294–307, Mar. 2005, doi: [10.1109/TIP.2004.838698](https://doi.org/10.1109/TIP.2004.838698).
- [12] F. Zhou et al., "Dual-granularity feature alignment for change detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4487–4497, 2025.
- [13] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–17, 2024, doi: [10.1109/TGRS.2024.3376673](https://doi.org/10.1109/TGRS.2024.3376673).
- [14] Z. Zhang, L. Bao, S. Xiang, G. Xie, and R. Gao, "B2CNet: A progressive change boundary-to-center refinement network for multitemporal remote sensing images change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11322–11338, 2024.

- [15] Z. Chen et al., "EDGE-Net: A building change detection method for high-resolution remote sensing imagery based on edge guidance and differential enhancement," *ISPRS J. Photogrammetry Remote Sens.*, vol. 191, pp. 203–222, Sep. 2022.
- [16] F. Wu, S. Dong, and X. Meng, "CFNet: Optimizing remote sensing change detection through content-aware enhancement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 14688–14704, 2025, doi: [10.1109/JSTARS.2025.3574173](https://doi.org/10.1109/JSTARS.2025.3574173).
- [17] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414, doi: [10.1109/TGRS.2024.3360516](https://doi.org/10.1109/TGRS.2024.3360516).
- [18] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.
- [19] X. Jia, Z. Chen, and S. Zhang, "CSGANet: Lightweight channel-split group attention for high-resolution remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 6907–6921, 2026.
- [20] J. H. Miao, S. Li, X. C. Bai, W. X. Gan, J. W. Wu, and X. H. Li, "RS-NormGAN: Enhancing change detection of multi-temporal optical remote sensing images through effective radiometric normalization," *ISPRS J. Photogrammetry Remote Sens.*, vol. 221, pp. 324–346, Mar. 2025.
- [21] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.
- [22] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [23] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 9992–10002.
- [24] B. Jiang et al., "VcT: Visual change transformer for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2005214.
- [25] X. Ma et al., "STNet: Spatial and temporal feature fusion network for change detection in remote sensing images," in *Proc. IEEE Int. Conf. Multimedia Expo.*, Jul. 2023, pp. 2195–2200.
- [26] C.-P. Chen, J.-W. Hsieh, P.-Y. Chen, Y.-K. Hsieh, and B.-S. Wang, "SARAS-Net: Scale and relation aware Siamese network for change detection," in *Proc. AAAI Conf. Artif. Intell.*, 2023 vol. 37, pp. 14187–14195.
- [27] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. 1st Conf. Lang. Model.*, Philadelphia, PA, USA, Oct. 2024.
- [28] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.
- [29] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CD-Mamba: Incorporating local clues into Mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016, doi: [10.1109/TGRS.2025.3545012](https://doi.org/10.1109/TGRS.2025.3545012).
- [30] D. Lu, P. Mausel, E. Brondizio, and E. Moran, "Change detection techniques," *Int. J. Remote Sens.*, vol. 25, no. 12, pp. 2365–2401, 2004.
- [31] S. Xie and Z. Tu, "Holistically-nested edge detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 1395–1403, doi: [10.1109/ICCV.2015.164](https://doi.org/10.1109/ICCV.2015.164).
- [32] T. Luo, Z. Ge, Q. Li, H. Dong, L. Bian, and C. Yang, "Adaptive edge-guided fusion network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 7305–7319, 2026.
- [33] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.
- [34] J. Tian, D. Peng, H. Guan, and H. Ding, "RACDNet: Resolution- and alignment-aware change detection network for optical remote sensing imagery," *Remote Sens.*, vol. 14, no. 18, 2022, Art. no. 4527, doi: [10.3390/rs14184527](https://doi.org/10.3390/rs14184527).
- [35] Y. Wang et al., "Lighter: A lightweight full-information and dual-guide network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 2924–2937, 2026, doi: [10.1109/JSTARS.2025.3647926](https://doi.org/10.1109/JSTARS.2025.3647926).
- [36] Z. Li, W. Zhou, H. Song, X. Zhang, and K. Zhang, "Details-enhanced multiscale network for building change detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 4530–4544, 2026.
- [37] A. Singh, "Digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, 1989.
- [38] W. Liu, J. Li, H. Wang, R. Tan, Y. Fu, and Q. Tian, "LCD-Net: A lightweight remote sensing change detection network combining feature fusion and gating mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 7769–7780, 2025, doi: [10.1109/JSTARS.2025.3544235](https://doi.org/10.1109/JSTARS.2025.3544235).
- [39] Z.-H. You, S.-B. Chen, J.-X. Wang, and B. Luo, "Robust feature aggregation network for lightweight and effective remote sensing image change detection," *ISPRS J. Photogrammetry Remote Sens.*, vol. 215, pp. 31–43, Sep. 2024.
- [40] L. Wang, W. Chen, H. Wang, and Z. Chen, "A novel Transformer-based multiscale Siamese framework for high-resolution remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 13087–13099, 2025, doi: [10.1109/JSTARS.2025.3569104](https://doi.org/10.1109/JSTARS.2025.3569104).
- [41] J. Dong, Z. Gao, and J. Li, "DCCANet: Dual-domain with collaborative cross-mapping attention network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 4418–4430, 2026.
- [42] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 9197–9206.
- [43] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, May 2020, Art. no. 1662, doi: [10.3390/rs12101662](https://doi.org/10.3390/rs12101662).
- [44] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multi-source building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.
- [45] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.
- [46] T. Xian et al., "A scale-temporal interaction network for remote sensing image change detection and a UAV-CD dataset," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Athens, Greece, 2024, pp. 8603–8607, doi: [10.1109/IGARSS53475.2024.10641915](https://doi.org/10.1109/IGARSS53475.2024.10641915).
- [47] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.
- [48] Z. Ying et al., "AEGL-Net: Adaptive multiscale global-local feature fusion network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5627219.
- [49] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.
- [50] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007, doi: [10.1109/TGRS.2006.885408](https://doi.org/10.1109/TGRS.2006.885408).
- [51] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 3431–3440, doi: [10.1109/CVPR.2015.7298965](https://doi.org/10.1109/CVPR.2015.7298965).
- [52] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 936–944, doi: [10.1109/CVPR.2017.106](https://doi.org/10.1109/CVPR.2017.106).
- [53] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in *Proc. Int. Conf. Learn. Representations*, 2016.
- [54] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7794–7803, doi: [10.1109/CVPR.2018.00813](https://doi.org/10.1109/CVPR.2018.00813).
- [55] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7132–7141, doi: [10.1109/CVPR.2018.00745](https://doi.org/10.1109/CVPR.2018.00745).
- [56] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19, doi: [10.1007/978-3-030-01234-2_1](https://doi.org/10.1007/978-3-030-01234-2_1).
- [57] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 4077–4087.
- [58] D. R. Martin, C. C. Fowlkes, D. Tal, and J. Malik, "A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2001, vol. 2, pp. 416–423, doi: [10.1109/ICCV.2001.937655](https://doi.org/10.1109/ICCV.2001.937655).

- [59] Z. Li et al., "Boosting remote sensing change detection via hard region mining," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5639812, doi: [10.1109/TGRS.2025.3598766](https://doi.org/10.1109/TGRS.2025.3598766).



Chaoyun Mai received the Ph.D. degree in information and communication engineering from Beihang University, Beijing, China, in 2017.

He is currently Associate Professor with the School of Electronics and Information Engineering, Wuyi University, Jiangmen, China. His current research interests include image processing and signal processing.



Haipeng He received the B.S. degree in computer science and technology from Guangdong Technology College, Zhaoqing, China, in 2024. He is currently working toward the master's degree in electronic information with the School of Electronics and Information Engineering, Wuyi University, Jiangmen, China.

His research interests include computer vision, object detection, and image processing.



Hao Xie received the Ph.D. degree in engineering from Sun Yat-sen University, Guangzhou, China, in 2022.

He is currently a Lecturer and Master's Supervisor with the School of Electronics and Information Engineering, Wuyi University, Jiangmen, China. His current research interests include digital image processing, deep learning, and multimedia content security.



Yikui Zhai (Senior Member, IEEE) received the Ph.D. degree in signal and information processing from Beihang University, Beijing, China, in 2013.

Since 2007, he has been working with Wuyi University, Jiangmen, China, where he is a Full Professor. He has been also Associate Dean with the School of Electronics and Information Engineering, Wuyi University, since 2021. He was a Visiting Scholar with the Department of Computer Science, the Università degli Studi di Milano, Italy, during 2016 to 2017, 2023, and 2024. His research interests include image

processing, deep learning, optical character recognition, object detection, UAV change detection, and self-supervised learning.



Zhiyuan Su received the B.S. degree in telecommunications engineering from Wuyi University, Jiangmen, China, in 2023, where he is currently working toward the master's degree in information and communication engineering with the School of Electronics and Information Engineering.

His research interests include intelligent information processing, remote sensing change detection, and image processing.



Panrong Chen received the B.S. degree in electronics and information engineering from Wuyi University, Jiangmen, China, in 2024, where he is currently working toward the master's degree in electronic information with the School of Electronics and Information Engineering.

His research interests include computer vision, object detection, and image processing.



Zhiping Peng received the Ph.D. degree in computer application technology from South China University of Technology, Guangzhou, China, in 2007.

He is currently the President and a Professor with Wuyi University, Jiangmen, China. He is a Postgraduate Supervisor and a recipient of the Special Government Allowance of the State Council. He has authored or coauthored nearly 100 papers, published one monograph, received seven provincial and ministerial science and technology awards, and been granted 29 invention patents. His research interests

include cloud computing, machine learning, intelligent optimization, and fault diagnosis.